



Sentient architecture promises insight into our evolving relationship with AI

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As the sun set on April 11, 2018, around 400 people gathered in Indiana University's Luddy Hall, anticipating a sort of birth. There, in the home of the School of Informatics, Computing, and Engineering, all eyes pointed to *Amatria*, a woven cloud of white Mylar and clear acrylic plastics, wire, glass, laser-cut stainless steel, and electronics hanging from the glass ceiling.

This new work of "sentient architecture" by multidisciplinary researcher and artist Philip Beesley of the University of Waterloo and colleagues hung still. Then Beesley reached up. Sensing his movement, *Amatria* shuddered. Vibrations near Beesley's hand rippled outward accompanied by bursts of warm light and sounds ranging from otherworldly clinks and bellows to natural gurgles and hisses.

Amatria is at once art installation, architectural prototype, and test bed for artificial intelligence. Beesley created the sculpture in collaboration with researchers at Indiana University, as well as electrical engineer Rob Gorbet of the University of Waterloo and other members of Beesley's Living Architecture Systems group (LAS), which includes an international team of 90 architects, scientists, engineers, and sound artists.

The goal of the LAS group is to make buildings come alive, revolutionizing built spaces and our relationships with them. The group aspires to create an architectural structure that is more integrated with the natural world, a metabolism that enables self-renewal, and an artificial intelligence capable of curiosity and even empathy.



Amatria, a new work of "sentient architecture" by Philip Beesley and others in LAS, is at once art installation, architectural prototype, and test bed for artificial intelligence. Image courtesy of Philip Beesley (photographer).



Nathanael Scheffler, part of the University of Waterloo's Living Architecture Systems group, was one of many students who helped piece together *Amatria*'s nearly half a million parts. Image courtesy of Andreas Bueckle (photographer).

LAS has exhibited more than 50 installations around the world. But *Amatria* is unique in that it functions as a permanent test bed in an academic community where it can evolve. As researchers collaborate with the sculpture's creators to add components and adjust *Amatria*'s code, they and visitors will work through many questions: How close can *Amatria* come to behaving like an actual living thing? And what does the piece suggest about our relationship with the artificially intelligent spaces we inhabit now and in years to come?

Curious Machine

Once Beesley animated *Amatria*, others began clapping and waving to draw their own responses. In some ways, this interaction is familiar. "The control systems that are being developed here can be understood as the next generation of active controls that have been with us since the first shopping center door opened in front of us," says Beesley. The difference is that *Amatria* not only responds but is driven to learn.

Amatria will soon be endowed with an artificial intelligence called the curiosity-based learning algorithm (CBLA). The algorithm will drive *Amatria* to search for examples of new behavior from itself and from people as they react with the sculpture's actuators, which generate light, sound, and vibrations. Starting in the fall, if a human responds to a particular actuator the same way every time, then the sculpture

should learn that response and come to "expect" it. The sculpture may then experiment with other actuators where other people are present. With additional sensors and a more developed CBLA, an even more tailored response may be possible.

Katy Börner, an Indiana University engineering and information science professor, expects the piece to evolve over time. "If *Amatria* has seen me come up the stairs many times in the morning," says Börner, alluding to the sculpture's future capabilities, "she is not going to take much attention away from other things she is interested in." Börner, who was instrumental in bringing *Amatria* to the school, prefers to refer to the sculpture as "she."

Modern-Day Quilting Bee

Beesley describes the sculpture's assembly with the technical skill of a scientist and the ease of a poet. His team, he explains, chained together vinegar cells—simple batteries that use vinegar as an electrolyte for conducting electricity. These vessels, which last about 3 months, will constantly stimulate *Amatria* with a weak electrical current, an energy that Beesley says is akin to "the impulse that creates breathing" or "the swell of the ocean."

Amatria would not be possible, he says, without not only sophisticated technology but a great many human hands. About 60 volunteers from the Bloomington community and schools across the university, as well as about 60 more volunteers back in Toronto, helped form and assemble the sculpture's nearly half a million parts in what Beesley likens to a quilting bee. "[*Amatria*] started out just packages of tiny, tiny pieces," says Sara Laughlin, a retired librarian who volunteered. "I love the fact that it is very organic," she says, alluding to the sculpture's form, which reminds her of underwater sea creatures. "And yet it is not at all organic. It is all plastic and electronic parts."

Much of *Amatria* was laser cut or 3D printed but still required a human touch. "I didn't expect something as intricate as this just from looking at little parts," says Sarah Lloyd, a freshman in the media program who helped cut out some of the 3D-printed pieces. Many of the clear plastic acrylic components started out as flat, laser-cut pieces, shaped like snowflakes cut from paper. Volunteers heated them in small ovens and then pulled the softened plastic up from the centers, transforming them into miniature latticed towers that function as structural components for the piece.

Ingredients of Life

The many pieces of *Amatria* form three main components—compartment, information, and metabolism—that Beesley says correspond to the "minimum ingredients of life."

The "compartment" gives the sculpture its shape and texture. White feathers cut from Mylar hang below the latticed acrylic towers, reinforced in places with stainless steel. The team optimized the geometric patterns of the towers to form minimal waste while still providing resilient, yet lightweight support. "It looks like you're inside of a neural network or a cell," says David

Ebbinghouse, a local artist and alumnus of the school's undergraduate fine arts program, who also volunteered.

Amatria's artificial intelligence fulfills the "information" component. The sculpture senses its surroundings through a series of smart microphones, which record the pitch and amplitude of sounds, as well as infrared sensors, which detect movement. *Amatria* also has internal microphones and electrical current sensors that give it feedback about its own actions and support its ability to learn. Microprocessors process input from the sensors and trigger actuators to respond by lighting bulbs, pumping sound through speakers, and generating vibrations with tiny motors.

"It is very inviting and it is also threatening. It is very familiar, but it is also alien."

—David Ebbinghouse

Amatria's "metabolism" is the least developed of the three components. Clear glass vessels hold, for example, copper and potassium salts that combine over time under osmotic pressure to form spherical, aqueous vesicles known as Traube cells that float between layers of heavier-than-water oil and lighter-than-water oil. Beesley and colleagues hope that these same sorts of cells could one day be used to form a self-renewing protective layer that clothes the outside of a building like ivy, possibly even playing a role in carbon capture and air filtering.

"Philip is constantly revising his systems to make them closer to living things," says neuroscientist Mark Cohen of the University of California, Los Angeles. During the conception of *Amatria*, Cohen contributed to the team's discussions of how the sculpture could mirror biological systems, such as having a segmented structure like annelids. But while Beesley likes to refer to his works as a sort of life, Cohen calls the pieces "biomimetic." "They are like living things," he says.

A Living Home

Amatria and Beesley's other works are architectural prototypes, but the group intends to one day form freestanding buildings. They envision that a structure such as the airy yet durable layers in *Amatria* could form a permeable exterior that protects inhabitants without blocking out the natural world.

Of course, people would have to decide whether they want to make a sentient space home. "[*Amatria*] is very contradictory," says Ebbinghouse. "It is very

inviting and it is also threatening. It is very familiar, but it is also alien."

"There is a degree of science fiction in it," says professor and architect Eric Jenkins of the Catholic University of America's School of Architecture and Planning, who helped arrange volunteers for the construction of *Sentient Chamber*, Beesley's November 2015 through May 2016 installation at the National Academy of Sciences in Washington, DC. "But architecture shares a great deal with science fiction." Because buildings last a long time, architects are always trying to envision the needs of inhabitants far into the future, he says.

There are multiple ways to view *Amatria*'s symphony of sound, light, and movement. Börner's PhD student, Andreas Bueckle, is developing an app that visualizes *Amatria*'s inner workings. At the unveiling, visitors reached toward infrared sensors as they watched a prototype of the app on a tablet. The screen showed a stream of virtual particles pouring from the sensors. As a hand got closer, the actual sculpture vibrated and the particles on the screen turned from white to red. "It adds a level of playfulness because not only is the sculpture going to react, the visualization is going to react too," says Bueckle. "I think it is very powerful for people to see their own data as they create it."

The same sort of app could visualize the sensors and actuators in the smart devices connected via the Internet, from televisions and coffee makers to thermostats and washing machines—all part of the much-heralded Internet of Things (IoT; see www.pnas.org/content/113/40/11059). If designers and artists understood these connections, says Börner, they might invent new IoT setups that are radically different and likely more desirable than what engineers and computer scientists would have ever envisioned.

On the day of the unveiling, small *Amatria*-inspired pieces of plastic sculpture and laser-cut feather lay strewn on office desks and classroom tables in Luddy Hall like fallen spider plant seedlings. Some, built during a summer camp, were towers that responded to an outstretched hand with a bowing feather. Others were feathered clips for hair or clothing.

The goal, says Börner, is eventually to have these *Amatria* offshoots communicate with the mother sculpture directly, bringing people and *Amatria* into more real-time and direct contact. As this relationship develops, she predicts we will learn more not only about AI but also ourselves. "The AI we create will reflect who humans are," says Börner, "just like children are a reflection of their parents."